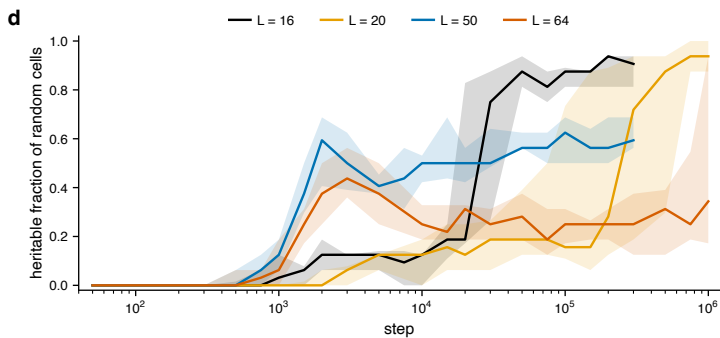
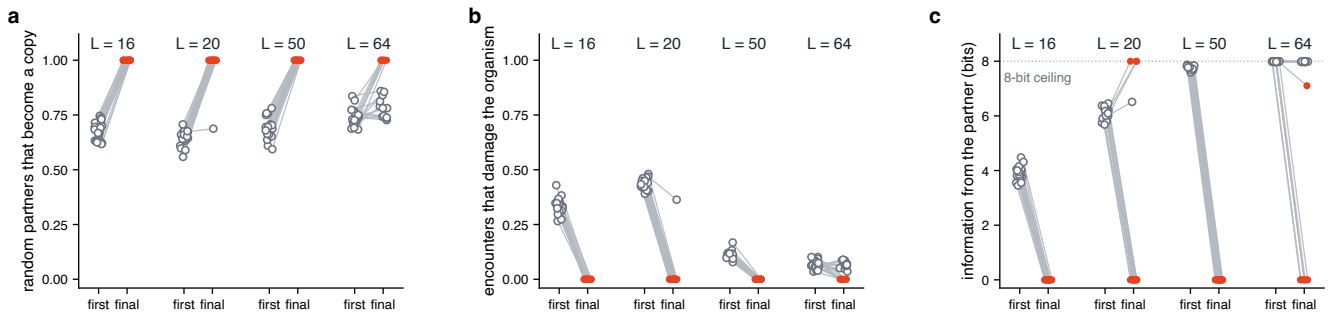


● loop instruction (jump, return or LDIR) ○ no loop instruction

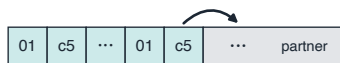


e

worlds of 20 by horizon	loop instruction	copies $\geq 95\%$ of partners	byte-identical to the modal tape
L = 16 300k steps	20	20	17
L = 20 1M steps	19	19	4
L = 50 300k steps	20	20	14
L = 64 1M steps	8	8	5

f

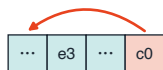
open · the first replicator



executes partner code

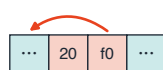
copies 0.68 of partners; damaged in 0.33 of encounters (medians, 20 worlds)

RET NZ · 16 bytes



its written bytes double as its return addresses

JR NZ · 50 bytes



a relative jump of 16 bytes back, through the address wrap

DJNZ · 20 bytes



a counted jump of 27 bytes back, through the address wrap

LDIR · 20 bytes



the block copy repeats in place

closed: copies 1.00 of partners · 0.00 self-damage

closure is a cycle in control flow, not a wall around the bytes